2513 char gen / 2716 eprom adapter issues

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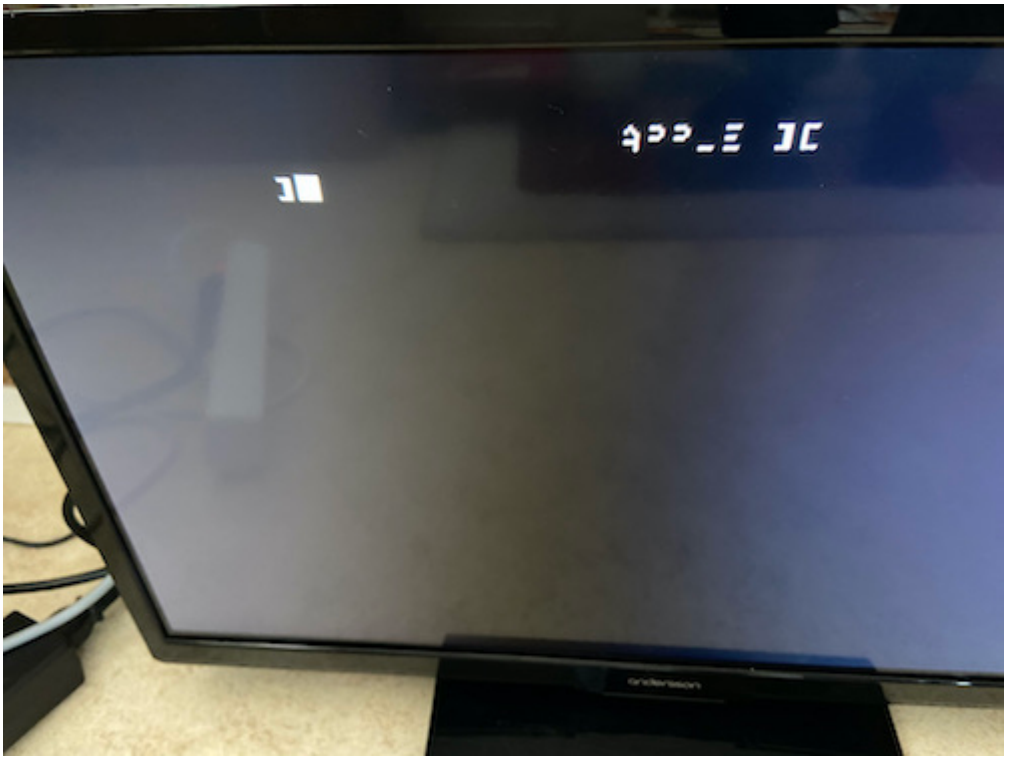
March 17, 2021 - 1:20pm #1

blindowl
Offline
Last seen: 4 years 1 month ago
Joined: Feb 2 2021 - 05:51
Posts: 21

2513 char gen / 2716 eprom adapter issues

Hi, I have a II+ rev. 4 with a Swedish 2513 character generator that I want to replace with an English character ROM. I've built a little adapter board so I can use 2716 EPROMs (or actually 28C16 EEPROMs), using this PCB as a template: https://oshpark.com/profiles/Benchoff

Everything works fine except the first column of pixels is always black:



According to the 2513 schematics it's **Out 5** that's not working correctly, but I've gone through all the connections several times with

the multimeter and can't really find any fault. When I put the 2513 back all characters are display correctly again, so it's definitely something wrong with my adapter board. Or am I missing something here regarding 2716/28C16 usage?

Any ideas?

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March 17, 2021 - 5:39pm #2

DeanClaxton

Offline

Last seen: 1 year 1 month ago

Joined: Aug 10 2006 - 22:21

Posts: 126

If you shoot me an email with

If you shoot me an email with your ROM image I can check that its ok (deanclaxton@gmail.com).

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March 18, 2021 - 12:11pm #3

blindowl

Offline

Last seen: 4 years 1 month ago

Joined: Feb 2 2021 - 05:51

Posts: 21

Thanks, that's kind of you,

Thanks, that's kind of you, but the ROM dumps I've tried are all downloaded from Asimov. They are most likely good.

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March 18, 2021 - 10:01pm #4

DeanClaxton

Offline

Last seen: 1 year 1 month ago

Joined: Aug 10 2006 - 22:21

I wonder if the bit alignment

I wonder if the bit alignment in the image you have is not suitable for 2513 - perhaps its an image for a 2316B replacement? Also would depend on how the adapter is wired. Take a look at the Rev 7 schematic vs pre-Rev 7.

Posts: 126

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March 19, 2021 - 1:17am **#5**

blindowl

Offline

Last seen: 4 years 1 month ago

Joined: Feb 2 2021 - 05:51

Posts: 21

DeanClaxton wrote:I wonder if

> ***DeanClaxton wrote:***
>
> *I wonder if the bit alignment in the image you have is not suitable for 2513 - perhaps its an image for a 2316B replacement? Also would depend on how the adapter is wired. Take a look at the Rev 7 schematic vs pre-Rev 7.*

Do you know how I can dump my existing 2513? If I can save it and re-program it to an eeprom I can verify if it shows the same behavior. I have an TL866II+ programmer and I'm not sure how to set it up for reading the 2513. Or if someone can point me to a know image of a 2513.

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March 19, 2021 - 7:26pm **#6**

DeanClaxton

Offline

Last seen: 1 year 1 month ago

Joined: Aug 10 2006 - 22:21

Posts: 126

Quick test - open your rom

Quick test - open your rom image in Omnivore, select all, Bytes -> rotate right. save it, burn it, test it :)

Or just open and have a look at the alignment of the leftmost pixel then see what the wiring on the adapter is doing.

For ROMX , our 2513 board set upgrades the capability of the pre rev 7 motherboard to that of the rev7 so we can use the same text rom images on both.

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March 21, 2021 - 12:10pm **#7**

blindowl

Offline

Last seen: 4 years 1 month ago

DeanClaxton wrote:Quick test

> ***DeanClaxton wrote:***
>
> *Quick test - open your rom image in Omnivore, select all, Bytes -> rotate right. save it, burn it, test it :)*

Joined: Feb 2 2021 - 05:51

Posts: 21

> *Or just open and have a look at the alignment of the leftmost pixel then see what the wiring on the adapter is doing.*
>
> *For ROMX , our 2513 board set upgrades the capability of the pre rev 7 motherboard to that of the rev7 so we can use*

Rotating the bytes did the trick! I wasn't aware of a difference between 2513 images and later dumps. I was also not aware of the Omnivore software, which seems very interesting to play around with! Thanks for pointing me in the right direction!

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March 21, 2021 - 4:44pm **#8**

DeanClaxton

Offline

Last seen: 1 year 1 month ago

Joined: Aug 10 2006 - 22:21

Posts: 126

Good stuff :) I"ve found

Good stuff :)

I"ve found Omnivore very useful for font manipulation - converting fonts for use on IIplus and IIe for example or from one format to the other. The bytewise functions are very handy and make it quick to do, plus you get to view the resulting bitmaps in the editor.

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March 22, 2021 - 2:14am **#9**

DeanClaxton

Offline

Last seen: 1 year 1 month ago

Joined: Aug 10 2006 - 22:21

Posts: 126

For future reference, the

For future reference, the Omnivore utility can be downloaded from here

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